

Emergency-expense resilience: a scoped household-resource reference

A compact piecewise-resource probability model predicts the SHED inability-to-pay 400 USD response better than matched controls on reserved states. The gain is modest and observational; it is not a causal poverty threshold, lending rule or demonstrated intervention.

Data and evaluation

2025 Federal Reserve SHED public-use cross-section, 12934 respondents. Forty-one states supply 10719 development respondents; ten complete hash-reserved states supply 2215 confirmation respondents. Final survey weights operate within equal state losses. Official source-coded labels map deterministically to values, with no imputation.

Target: **Unable to pay hypothetical 400 USD emergency expense by any listed means**, measured as binary EF3_h. Primary error: squared probability (Brier). Contemporaneous 2025 household survey association. No future distress observation and no income intervention.

Respondents nested in whole states; source respondents may share unreported household/environmental dependencies. States not independent populations. Development states only; five whole-state hash folds. No held-state outcomes/calibration. EF3 other response checkboxes and EF1 savings coverage excluded to avoid near-endpoint restatement.

Findings and equations

$$p = \text{logistic}(a + b_L(3 - R)_+ + b_H(R - 3)_+ + cS + dA + eA^2)$$

The notation and all fitted coefficients are defined below. Here logistic denotes the inverse-logit function; a positive-part subscript denotes $\max(x, 0)$.

P100-015-F01 - validated_extension (partially_supported). Piecewise resource and age/work associations yield a compact regional-transfer predictor of the reported inability-to-pay response.

$$p = \text{logistic}(a + b_L(3 - R)_+ + b_H(R - 3)_+ + cS + dA + eA^2)$$

A saturation-shaped resource association is consistent with heterogeneous liquidity constraints, but proxies/topcodes and overlapping self-reports preclude identifying a physical/economic threshold.

Falsifying evidence and limits: Very small margin over flexible comparator; source-aware cross-section and only ten held states; endpoint is hypothetical response, not observed emergency outcome.

P100-015-F02 - informative_falsification (supported). Removing work status or freeing household equivalence scaling did not improve the development-selected threshold model.

Attempt 005 removes S; 006 replaces resource hinge with independent logI/logH coefficients.

Independent predictive contribution does not establish causal unemployment risk; S includes retirement and voluntary nonparticipation.

Falsifying evidence and limits: Attempt 005 is retrospectively better on confirmation mean; it was not promoted or retuned, showing model-selection uncertainty.

Selected executable model: attempt_004. Let $R = \ln(I/\sqrt{H})$, where I is the declared household-income bracket proxy in thousand USD/year and H household size; $A = (\text{age} - 45)/20$; $S = 1$ for no paid/profit work last month. Selected $p = \text{logistic}(a + b_L \cdot \max(3 - R, 0) + b_H \cdot \max(R - 3, 0) + c \cdot S + d \cdot A + e \cdot A^2)$. Coefficients in that order are [-0.8836098559778368, 0.31380714954933303, -1.6092569888184314, 0.8557459418933252,

-0.3187511278524709, -0.3612550550290176]. The fixed R=3 hinge is a model diagnostic, not an official poverty or acceptance threshold.

All coefficients, input definitions, training groups and transformations are in `rules.json`; the runnable reference performs no fitting.

Compact numerical evidence

Errors are computed within declared complete groups (with final survey weights when supplied), then averaged equally across groups.

Frozen model	Development error	Confirmation error	Worst confirmation group
reference	0.08374207	0.07371215	0.1212739
constant	0.1042601	0.08555106	0.1522752
domain	0.08852593	0.07583128	0.1289653
flexible	0.08519179	0.07404488	0.1254853

Confirmation contains 2215 scored observations in 10 groups. Individual-group scores and every attempted model remain available. Reserved-state Brier 0.073712 versus constant 0.085551, simple resource 0.075831 and flexible 0.074045. This is only about 0.45 percent relative better than the flexible comparator; no decisive population superiority claimed. Worst-state reference 0.121274 versus flexible 0.125485. Attempt 005 has a lower confirmation mean but was not development-selected and remains diagnostic.

Interpretation, limitations and use

Positive-weight complete cases, state-balanced within-state survey-weighted predictive risk; not national prevalence or causal financial access. Income proxy upper bracket 175 k is a modeling convention. D1A0 reflects no paid work, not unemployment or a job-loss shock. Whole-group errors/ranges; no row bootstrap or manufactured independent-population confidence. Confirmation becomes exposed after freeze.

Useful for evaluating grouped, weighted survey-risk modeling and whether compact interpretable equations can match flexible models. No causal effect of employment, actual household income amount or achieved financial-welfare benefit is demonstrated. The question measures inability to pay by any listed means, not inability to pay using cash alone.

Reproducibility and prior art

Run `python run.py` to verify hashes and replay every frozen equation. Predictions use declared input columns only. The shared benchmark evaluator can score alternative equations under the same task; numerical agreement with these coefficients is not required. Public source-aware confirmation is now exposed. No independent experimental replication or certified novelty is claimed.

- Official SHED public data/codebook; emergency expense resilience is a published descriptive endpoint. Resource equivalence and age/life-cycle associations are prior model families, not established causal mechanisms in this snapshot. (checked 2026-10-02).